sesh-meeting-2026-02-04

Ian Chi

February 12, 2026

given n coins, each with probability p of flipping heads, what is the probability that all of them flip heads. this is equal to $(p)^n$.

(in this case, p is analogous to all x_n expected values being under 2/3, sampling each x_n m times, given m is sufficiently large, what is the probability that one sample is above 2/3. this is the a light detection case. p is thus close to 1.)

if i want to use union bound, the union of all events being 1 is less than the sum. each sample goes from p probability to $\frac{n+1}{n}p$. it requires n extra samples by chernoff? since 2/3 is fixed and p, the expected value is known, we can use multiplicative percentage error

I want $(p)^n > 2/3$. 1 extra sample requires p to increase by α. to find α,

$$\Delta p = \left(\frac{2}{3}\right)^{\frac{1}{n+1}} - \left(\frac{2}{3}\right)^{\frac{1}{n}} = -\frac{2}{3}^{1/n}$$

$$\frac{2}{3} = (1-p)^n \tag{1}$$